

1. Administrative & Study Identification

Full Study Title: A Phase 3, Multicenter, Randomized, Partially Blinded, Palivizumab-Controlled Study to Evaluate the Safety, Efficacy, and Pharmacokinetics of MK-1654 in Infants and Children at Increased Risk for Severe RSV Disease **Study Title:** Clesrovimab (MK-1654) in Infants and Children at Increased Risk for Severe Respiratory Syncytial Virus (RSV) Disease (MK-1654-007) **Short Title/Acronym:** SMART Study (MK-1654-007) **Protocol Number:** MK-1654-007 **NCT Number:** NCT04938830 **Posting ID:** 145584772246977579 **Trial Status:** COMPLETED (Study has been completed) **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Clesrovimab (MK-1654 / Enflonsia™) **Active Comparator:** Palivizumab (Trade Name: Synagis | ATC Code: J06BD01) **Phase of Development:** Phase 3 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of clesrovimab compared to palivizumab as assessed by the proportion of participants experiencing adverse events (AEs). **Secondary Objectives:** To evaluate the efficacy (reduction in the incidence of RSV-associated medically attended lower respiratory infections [MALRI] and RSV-associated hospitalizations) and the pharmacokinetics (monoclonal antibody serum concentrations) of clesrovimab compared to palivizumab over two RSV seasons.

2.2 Background & Rationale Respiratory syncytial virus (RSV) is a common virus that can cause severe, sometimes life-threatening illness in infants and children born prematurely or with certain heart and lung conditions. This study aims to address the need for a prophylactic option that can prevent RSV lower respiratory tract disease with a single, extended half-life dose. Clesrovimab is designed to provide direct, rapid, and durable passive immunization through 5 months (a typical RSV season) with a single injection, reducing the burden of the standard monthly palivizumab injections currently required for high-risk infants.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Palivizumab-controlled) **Blinding:** Partially Blinded / Double-blind **Randomization:** 1:1 (clesrovimab vs. palivizumab)

3.2 Planned Enrollment & Duration Target **\$N\$:** 1,003 infants/children **Number of Sites:** Multicenter global study (~147 locations) **Estimated Study Start:** 30-Nov-2021 **Estimated Primary Completion:** 29-Apr-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Infants from birth up to 1 year of age (entering their first RSV season) and children under 2 years of age (entering their second RSV season). 2. Participants at increased risk for severe RSV infection recommended to receive palivizumab in accordance with national or local guidelines or professional society recommendations (e.g., prematurity ≤ 35 weeks gestational age, chronic lung disease [CLD] of prematurity, or hemodynamically significant congenital heart disease [CHD]). 3. Is available to complete the follow-up period.

4.2 Exclusion Criteria 1. Requires mechanical ventilation at the time of enrollment. 2. Has a life expectancy < 6 months. 3. Has known hepatic or renal dysfunction, or chronic seizure disorder. 4. Is hospitalized at the time of randomization unless discharge is expected within 7 days after randomization. 5. Has severe immunodeficiency, is severely immunocompromised, or has known hypersensitivity to any component of clesrovimab or palivizumab.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single fixed dose of 105 mg (0.7 mL) for the first RSV season. Eligible patients participating in a second RSV season receive an additional 210-mg dose. **Route of Administration:** Intramuscular (IM) injection. **Duration of Treatment:** Single administration providing protection through Day 150 (5 months) per season.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant therapies for underlying prematurity, CLD, or CHD. **Prohibited:** Use of other experimental RSV vaccines or preventive monoclonal antibodies outside of the study assignment. *Note: The active comparator group receives monthly intramuscular palivizumab (Synagis, ATC Code: J06BD01) per standard of care.*

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening / Baseline	Day 0	Informed Consent, Medical History, Randomization, First IM Injection
Interim (Season 1)	Up to Day 150	Safety assessments, AE monitoring, blood samples for PK, nose swabs (if symptomatic). Approx. 6 to 8 visits during Year 1.
Interim (Season 2)	Pre-Season 2	Second dose administration (210-mg clesrovimab)

for eligible patients. | | **End of Study** | Month 12 to 18 | Final physical examination, Safety & PK follow-up, Exit Interview. Approx. 10 to 12 total visits for Season 2 participants. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: Safety and tolerability of clesrovimab compared to palivizumab over the study duration, assessed by: 1. **Participants with solicited injection-site AEs in RSV Season 1:** Percentage of participants with solicited injection-site AEs in RSV Season 1. 2. **Participants with solicited daily body temperature with fever in RSV Season 1:** Percentage of participants with solicited daily body temperature with fever in RSV Season 1. 3. **Participants with solicited systemic AEs in RSV Season 1:** Percentage of participants with solicited systemic AEs in RSV Season 1.

7.2 Secondary & Exploratory Endpoints 1. **Participants with RSV-associated medically attended lower respiratory infection (MALRI) in RSV Season 1** (requiring ≥ 1 indicator of LRI or severity compared to palivizumab through Day 150). 2. **Participants with RSV-associated hospitalization in RSV Season 1** (through Day 150). 3. **Participants with solicited injection-site AEs in RSV Season 1.** 4. Monoclonal antibody (mAb) serum concentrations (Pharmacokinetics).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of common anticipated events including injection-site erythema, swelling, and pain, as well as systemic symptoms like irritability, somnolence, decreased appetite, rash, and fever. **Serious Adverse Events (SAEs):** Continuously monitored. Interim data noted a safety profile comparable to palivizumab with no drug-related serious AEs reported. **Data Monitoring Committee (DMC):** Internal and external safety monitoring committees oversee AE/SAE collection.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy bounds; Safety Set for all participants who received at least one dose of the study medication. **Sample Size Justification:** Target enrollment of $N = 1,003$ to ensure sufficient power to demonstrate non-inferiority or superiority in safety and efficacy (MALRI reduction) compared to the active control palivizumab. **Handling of Missing Data:** Handled via multiple imputation where applicable; exact time-to-event outcomes analyzed via Kaplan-Meier estimates and Cox proportional hazards models through 150 days post-dose.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to ensure protocol adherence, accurate AE reporting, and proper administration of IM therapies. **Audit Trail:** Complete eCRF tracking, secure centralized laboratory evaluations for PK blood samples and RSV confirmation (e.g., PCR testing on nose swabs).

11. Study Identifiers & Governance

Primary ID: MK-1654-007 **Registry Number:** NCT04938830 **Posting ID:** 145584772246977579 **Sponsor/Lead Organization:** Merck **Company:** Merck **Study Title:** Clesrovimab (MK-1654) in Infants and Children at Increased Risk for Severe Respiratory Syncytial Virus (RSV) Disease (MK-1654-007) **Official Regulatory Title:** A Phase 3, Multicenter, Randomized, Partially Blinded, Palivizumab-Controlled Study to Evaluate the Safety, Efficacy, and Pharmacokinetics of MK-1654 in Infants and Children at Increased Risk for Severe RSV Disease

12. Clinical Framework (RSV Specific)

Primary Condition / Disease Area: Respiratory Syncytial Virus (RSV) Infection **Preferred UMLS Name:** Respiratory Syncytial Virus Infections **MeSH Code:** D018357 **SNOMED ID:** 55735004 **Diagnosis Threshold:** Increased risk for severe RSV disease (e.g., prematurity, CLD, CHD) **Medical Methodology:** Passive Immunization (Monoclonal Antibody) **Optimization Target:** Prevention of RSV-associated MALRI and hospitalization

13. Study Procedures & Methodology

Management Pathway: Prophylactic administration prior to or during the RSV season. **Education Protocol (HCP):** Training on proper IM administration and AE tracking (differentiating typical injection site reactions from active infection). **Education Protocol (Patient/Caregiver):** Symptom diary training to recognize signs of MALRI (e.g., wheezing, retractions, hypoxemia, tachypnea). **Quality Audit Mechanism:** Laboratory verification of RSV infection and continuous mAb serum concentration profiling.

14. Observation & Follow-Up Schedule

Total Duration: Up to 18 months for participants enrolled in Season 2 **Onsite Visit Cadence:** 6 to 8 times during Year 1; up to 10-12 times total for 18-month participants **Remote Monitoring**

Frequency: Intermittent phone or digital diary follow-ups for AE tracking. **Checklist Review Interval:** Regular review of symptom indicators prior to clinical diagnosis of MALRI.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					